Sinusoidal Signals: Examples

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Key Concepts

1) Different musical notes can be obtained by varying the frequency of a sinusoid.

2) The frequencies of consecutive notes in western music are a multiple of $2^{1/12}$ because there are 12 notes in each octave.

3) A complex periodic signal like a note played by a saxophone can be approximated as a weighted sum of sinusoids whose frequencies are integer multiples of the fundamental frequency.

Sinusoidal Signals: Examples

Examples

- Sinusoids as tones in a scale

  Music theory

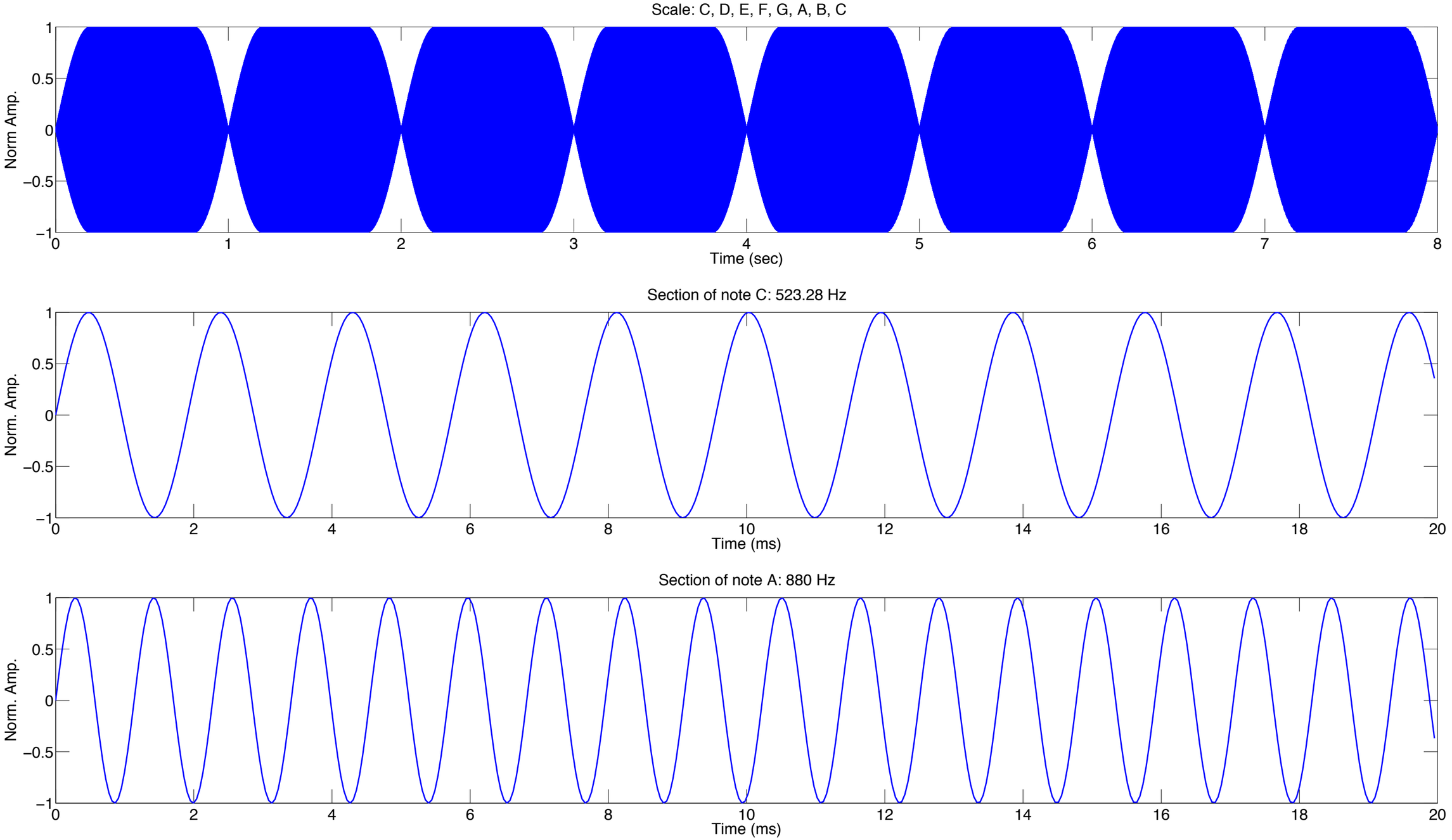
  a. 12 notes per octave or doubling of frequency

  b. Frequencies spaced by multiples of 2^(1/12)

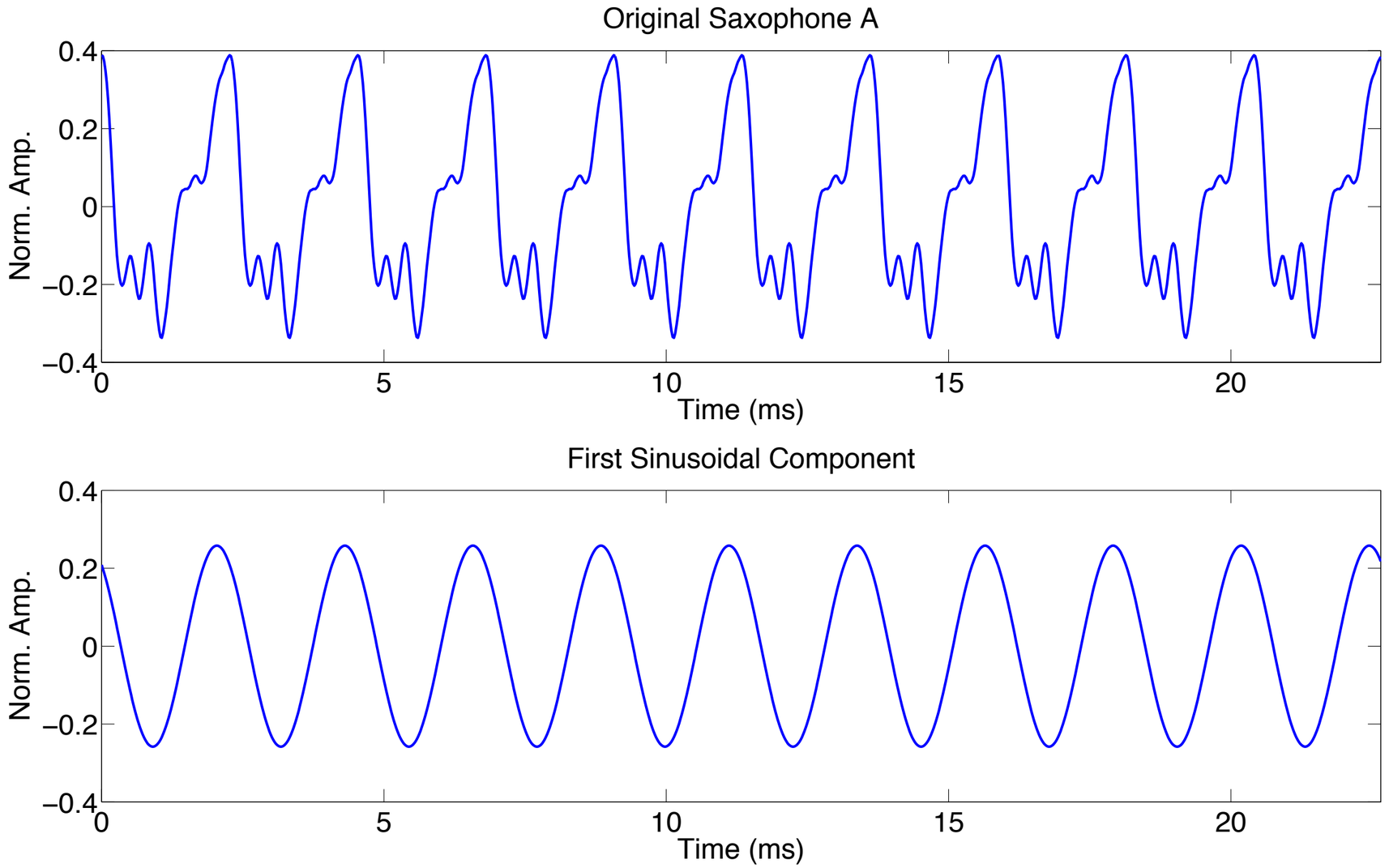
  c. C-major scale:

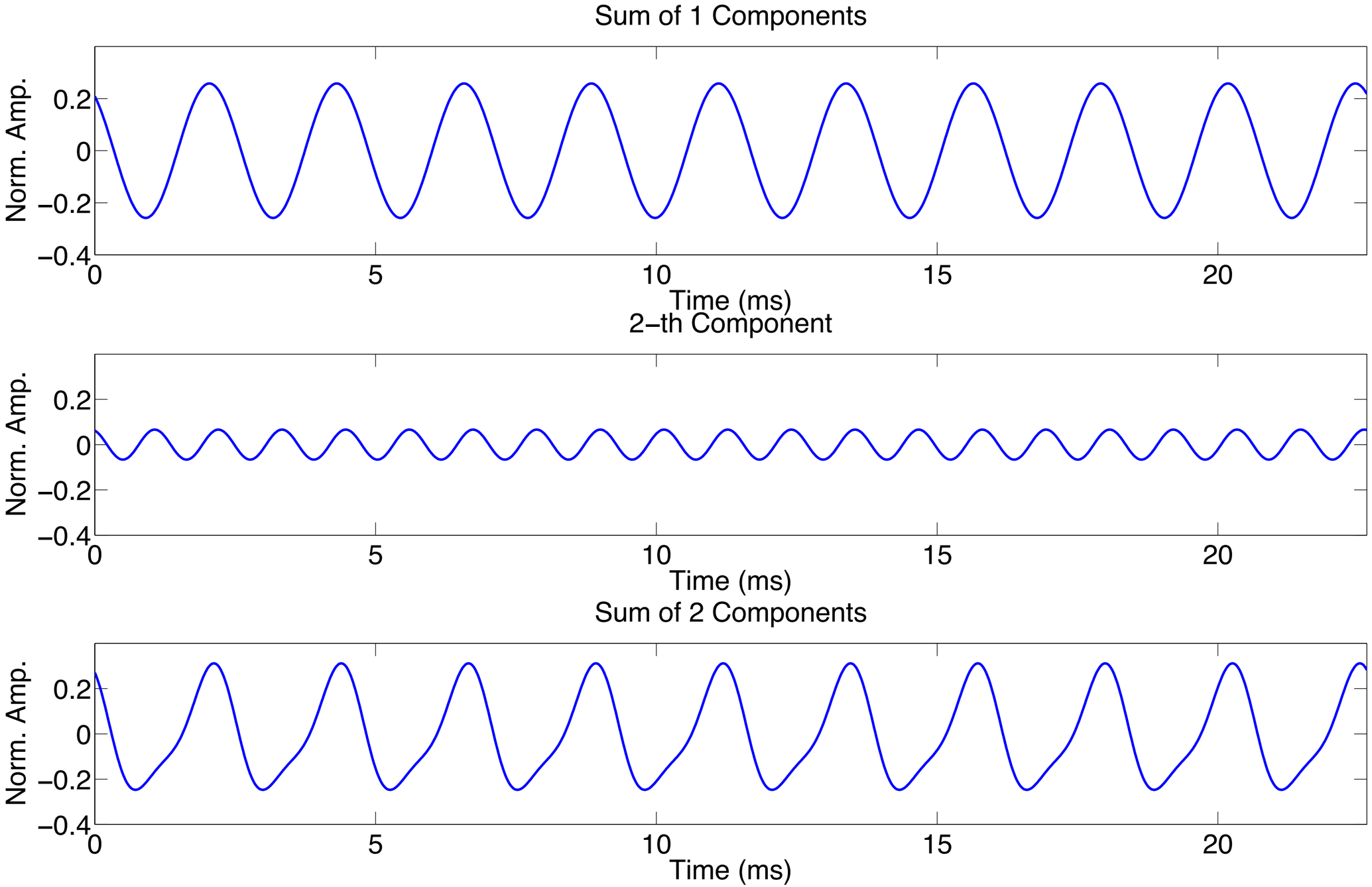
     C 523.28; D 587.36; E 659.28; F 698.48; G 784; A 880; B 987.84 Hz

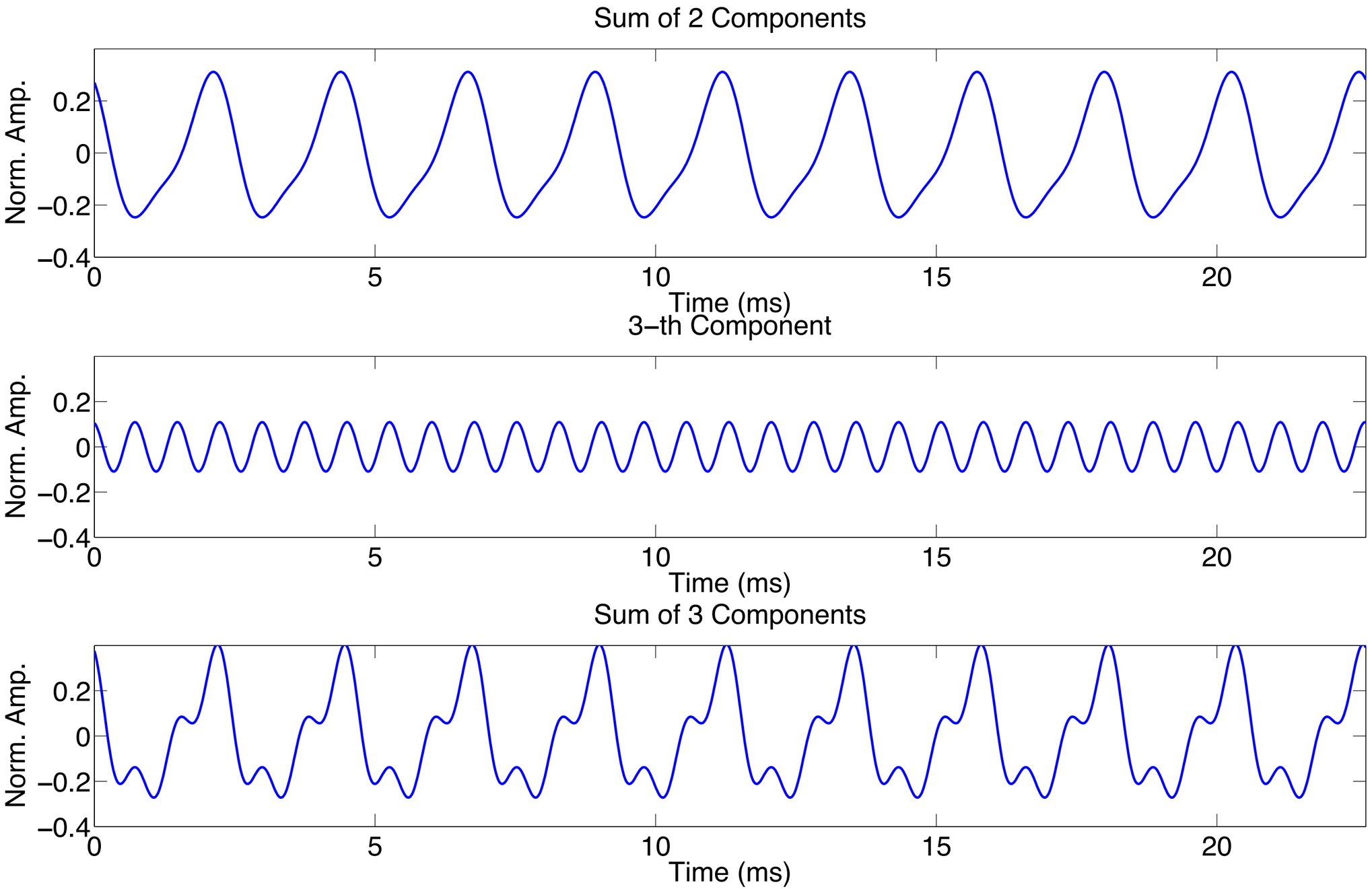
- Sum of sinusoids with different frequencies - Saxophone note A

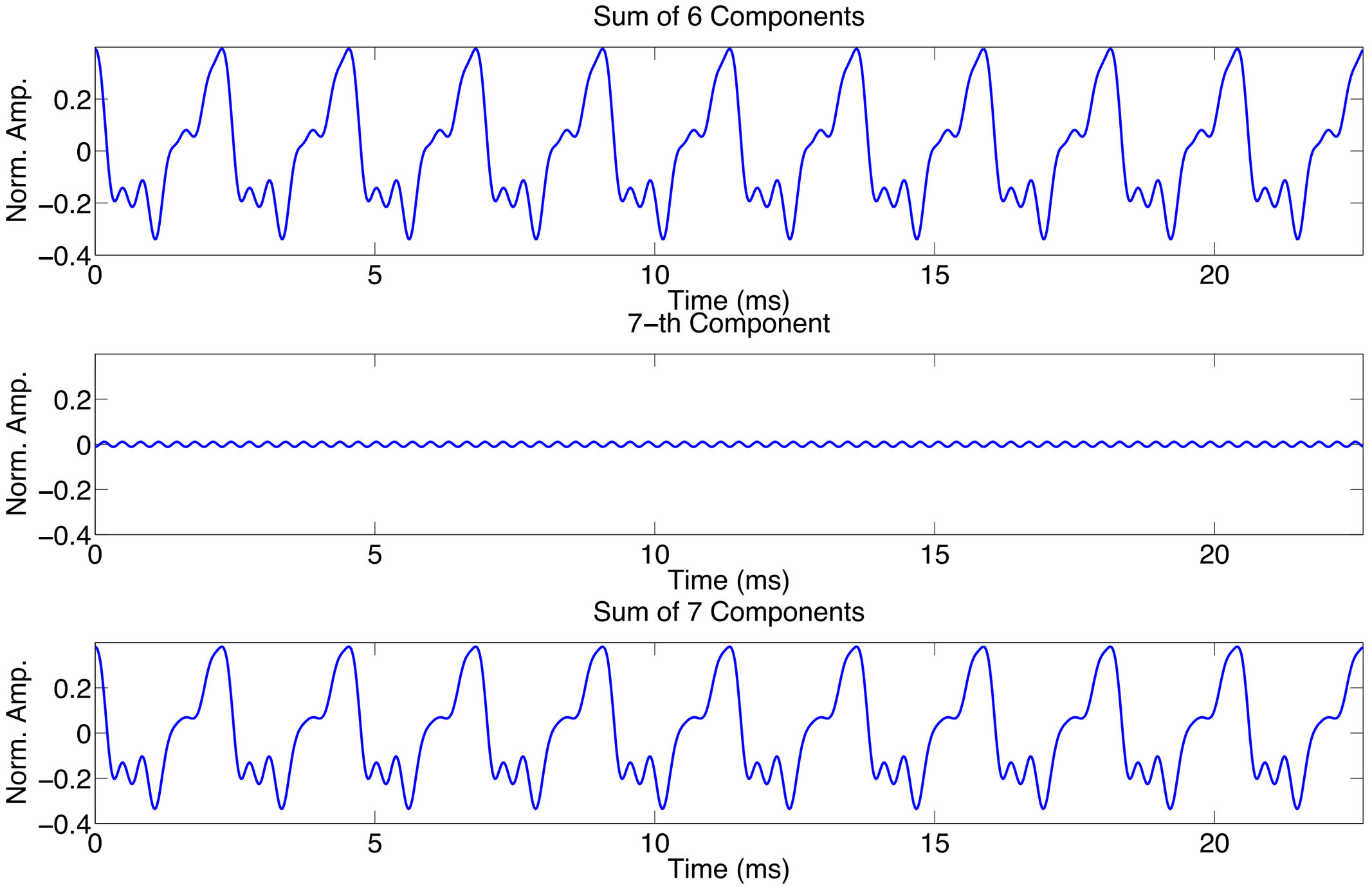

Scale: C, D, E, F, G, A, B, C
Norm Amp.
Time (sec)
Section of note C: 523.28 Hz
Norm. Amp.
Time (ms)
Section of note A: 880 Hz
Norm. Amp.
Time (ms)

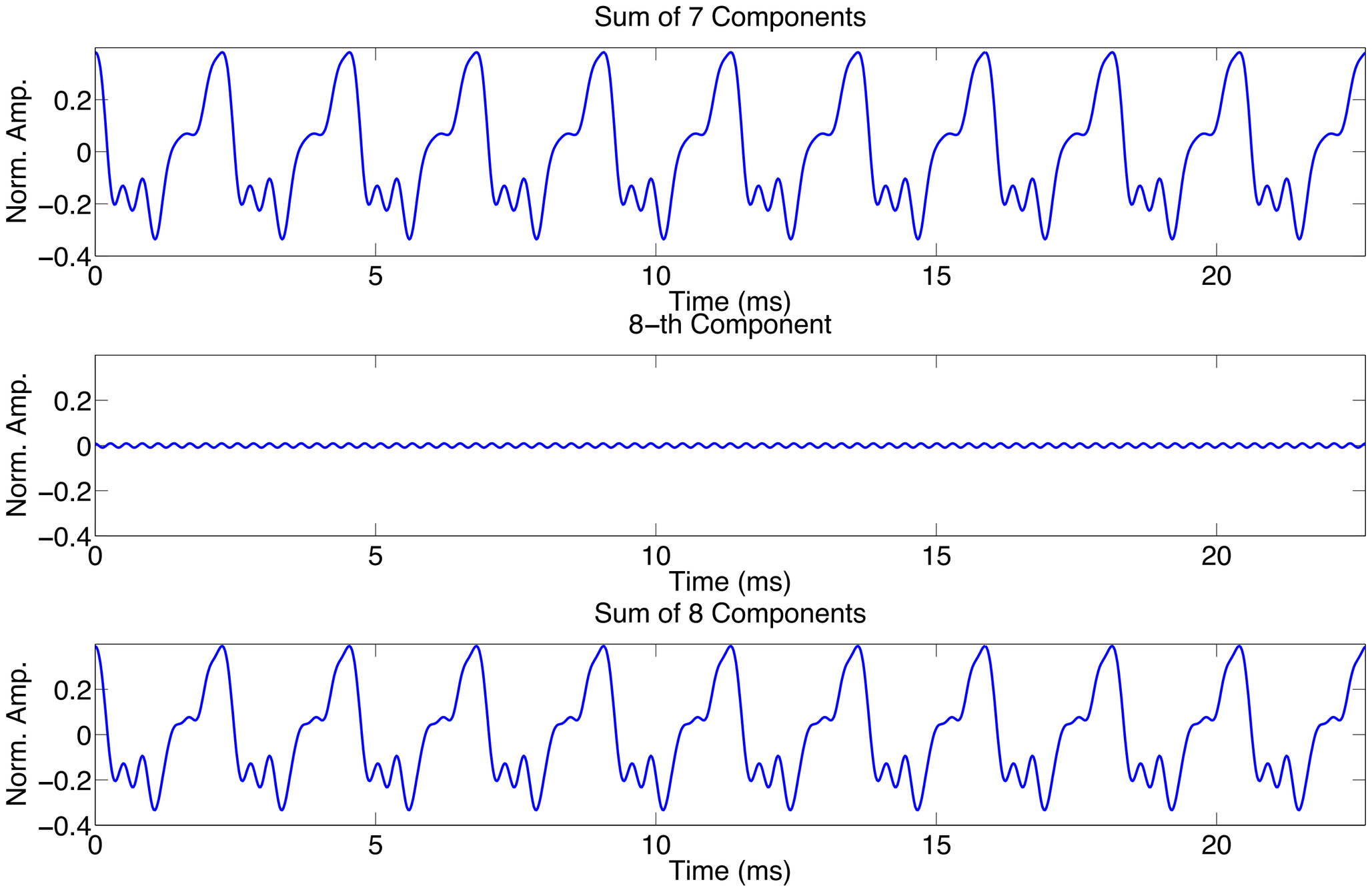
Saxophone Note A

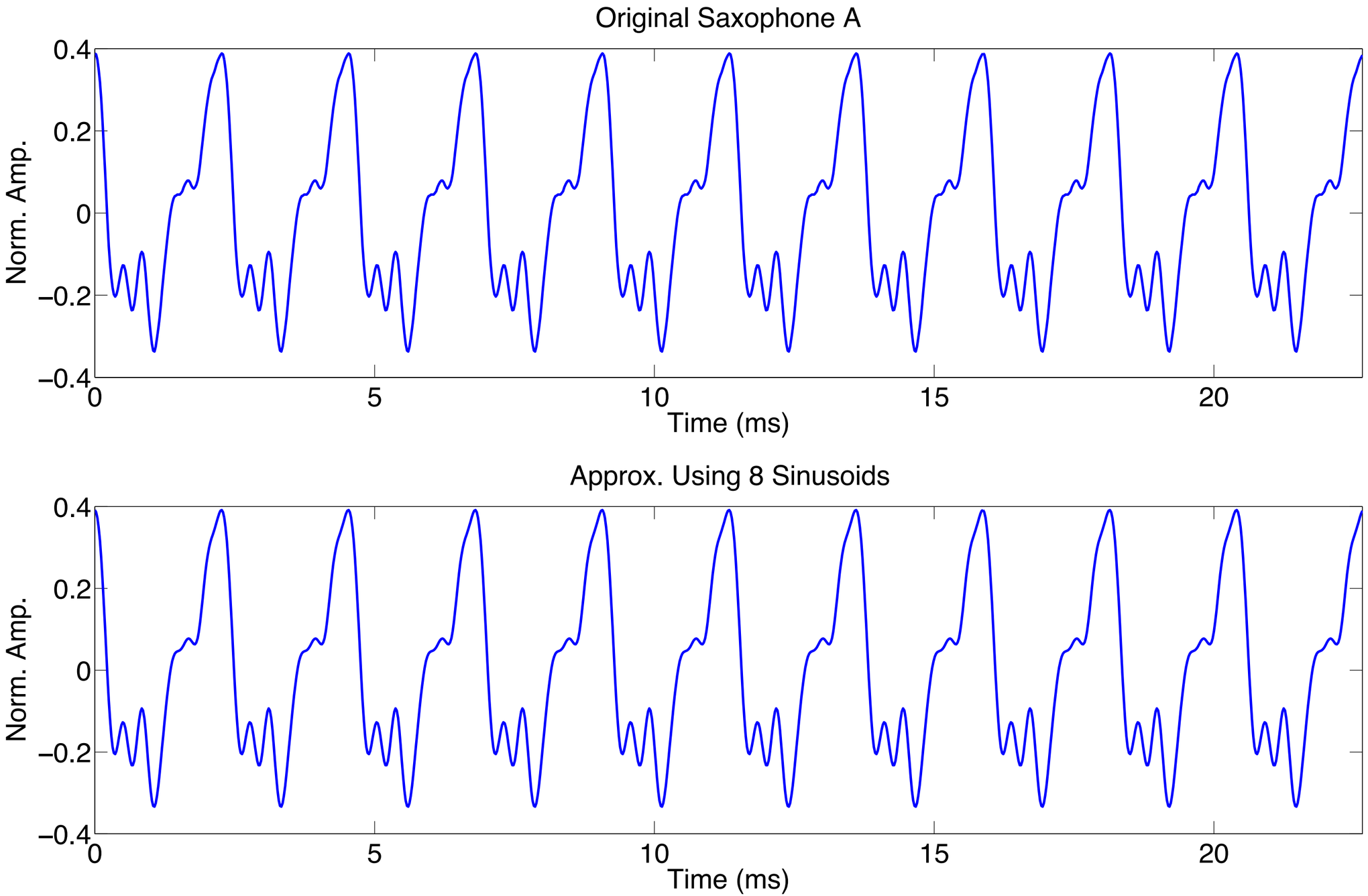



Sum of 1 Components
Norm. Amp.
0.2
0
−0.2
−0.4
0
5
10
15
20
Time (ms)
2−th Component
Norm. Amp.
0.2
0
−0.2
−0.4
0
5
10
15
20
Time (ms)
Sum of 2 Components
Norm. Amp.
0.2
0
−0.2
−0.4
0
5
10
15
20
Time (ms)


Sum of 2 Components
Norm. Amp.
0.2
0
−0.2
−0.4
0
5
10
15
20
Time (ms)
3−th Component
Norm. Amp.
0.2
0
−0.2
−0.4
0
5
10
15
20
Time (ms)
Sum of 3 Components
Norm. Amp.
0.2
0
−0.2
−0.4
0
5
10
15
20
Time (ms)


Sum of 6 Components
Norm. Amp.
0.2
0
−0.2
−0.4
0
5
10
15
20
Time (ms)
7−th Component
Norm. Amp.
0.2
0
−0.2
−0.4
0
5
10
15
20
Time (ms)
Sum of 7 Components
Norm. Amp.
0.2
0
−0.2
−0.4
0
5
10
15
20
Time (ms)


Sum of 7 Components
Norm. Amp.
0.2
0
−0.2
−0.4
0
5
10
15
20
Time (ms)
8−th Component
Norm. Amp.
0.2
0
−0.2
−0.4
0
5
10
15
20
Time (ms)
Sum of 8 Components
Norm. Amp.
0.2
0
−0.2
−0.4
0
5
10
15
20
Time (ms)


Original Saxophone A
Norm. Amp.
0.4
0.2
0
−0.2
−0.4
0
5
10
15
20
Time (ms)
Approx. Using 8 Sinusoids
Norm. Amp.
0.4
0.2
0
−0.2
−0.4
0
5
10
15
20
Time (ms)

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